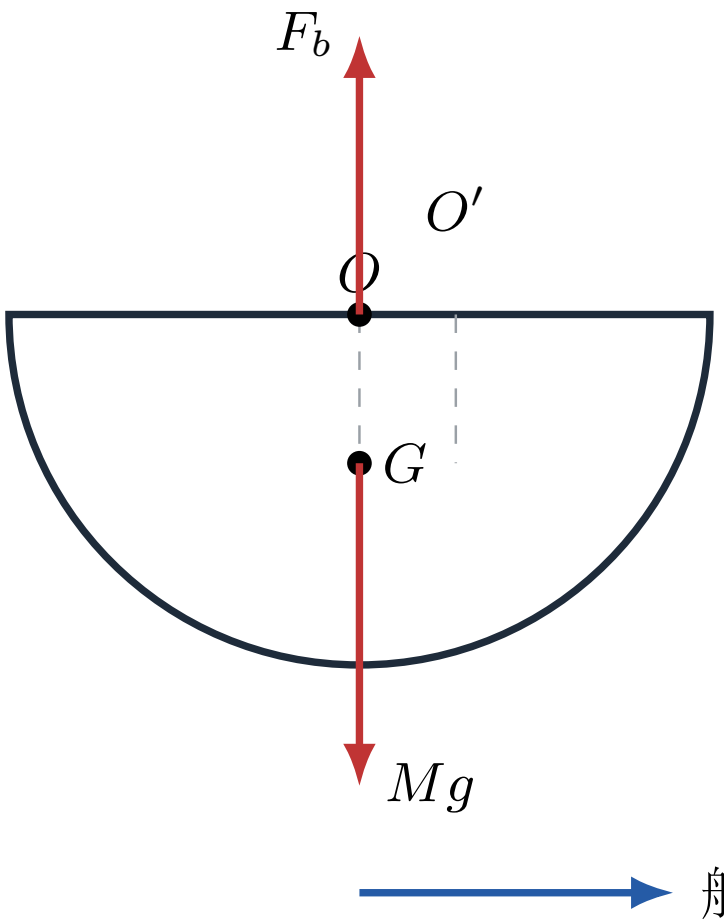


O 非固定铰链；G 的横坐标近似固定

$$I_G = I_O - Mh^2$$

$$T_2 = 2\pi\sqrt{I_G/(Mgh)}$$

$$\Delta\theta_{\text{cycle}} = 4mrI_O/(MhI_G)$$

快跑既改变平衡角，也产生姿态跳变